



A Paradigm Shift in Agentic RAG

How to combine advanced reasoning with RAG (arXiv)

Reasoning Workflow Types



Reasoning

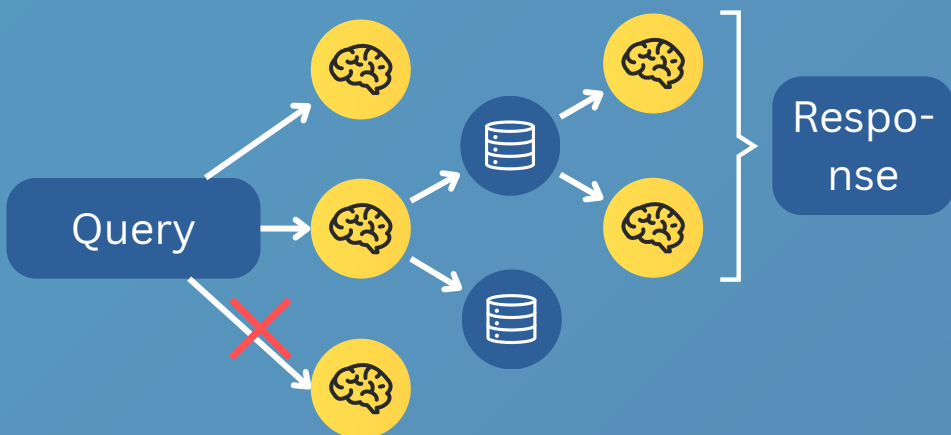


RAG

Chain-Based



Tree-Based



Example techniques:

- Tree-of-Thought (ToT)
- Monte Carlo Tree Search (MCTS)

Graph-Based



Example techniques:

- Walk-on-Graph
- Think-on-Graph

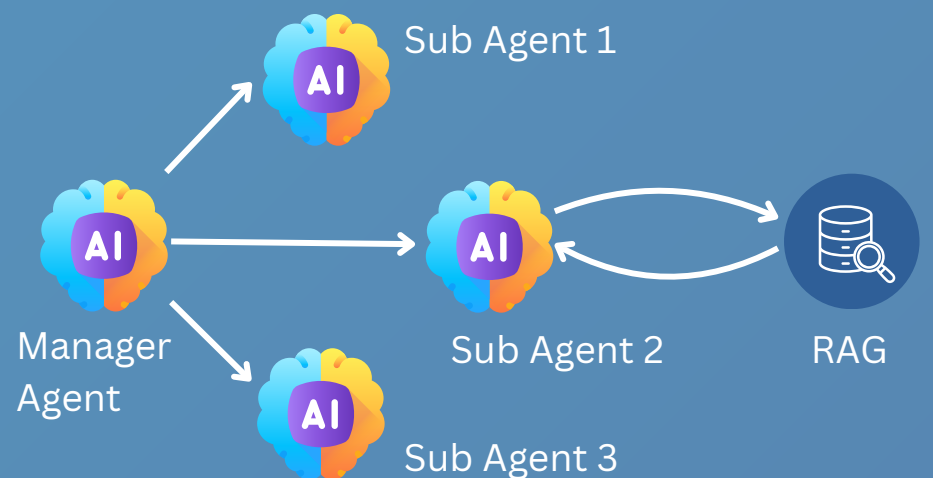
Agent Orchestration Types

Single Agent



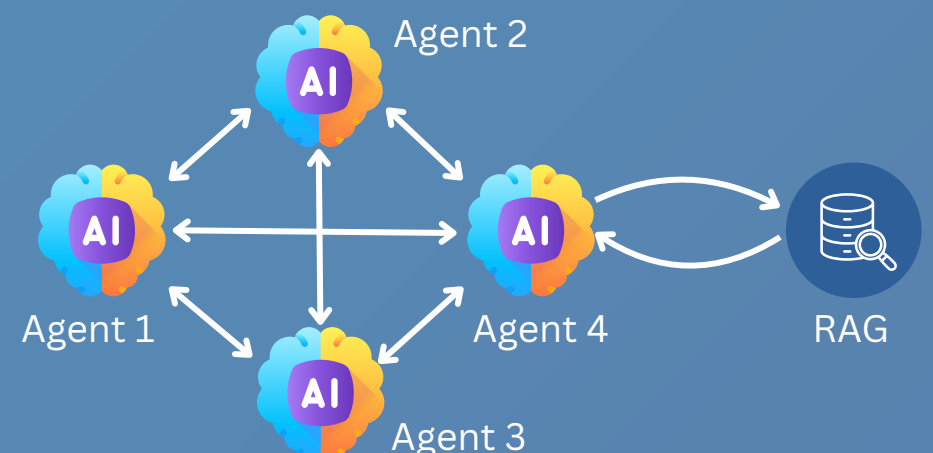
Example techniques: Prompting (ReAct, Search-O1), Supervised Fine-Tuning, Reinforcement Learning

Centralized Multi-Agent



Example techniques: HM-RAG (hierarchical multi-agent multimodal RAG), Chain of Agents

Decentralized Multi-Agent



Research: M-RAG (multiple partitions), MDocAgent (multimodal for doc. understanding)